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LAB REPORT

Course Code : CSE-4878
Course Title : Machine Learning and Data Mining Lab

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Remarks

1. Introduction

A traditional Security Operations Center (SOC) produces many alerts, and a human analyst has to decide which alerts are related, how serious they are, and what response is needed. This project, called AutoSOC, automates part of that work. A Random Forest classifies network events, and rule-based analysis then groups related malicious events into candidate multi-stage campaigns, scores their risk, and recommends a response. The correlated sequences are treated as candidates, not confirmed attack chains, and are checked against ground-truth data afterwards.

2. Objectives

- Train a model to classify normal and malicious network traffic and filter uncertain alerts by confidence.
- Correlate related events using network entities and time to find candidate multi-stage campaigns.
- Add MITRE ATT&CK context, risk scoring, incident investigation, and response recommendation.
- Validate the candidate campaigns separately using ground-truth data.
- Compare the assigned pipeline with the updated pipeline.

3. Dataset

The project uses TON_IoT network data. One labeled file (train_test_network.csv) is used for model training and held-out testing. Separate timestamped files (Network_dataset_1, 9, 16 and 18) are used for external evaluation and correlation, and the matching GroundTruth_Network files are used only after correlation for validation.

Item	Value
Original training dataset	211,043 rows, 44 columns
Exact duplicate rows removed	20,569
Rows after duplicate removal	190,474
Training samples	152,379
Testing samples	38,095

3.1 Attack Classes

The dataset has ten classes: normal and nine attack types. The classes are not balanced; MITM has only 1,041 samples, while normal has 42,040.

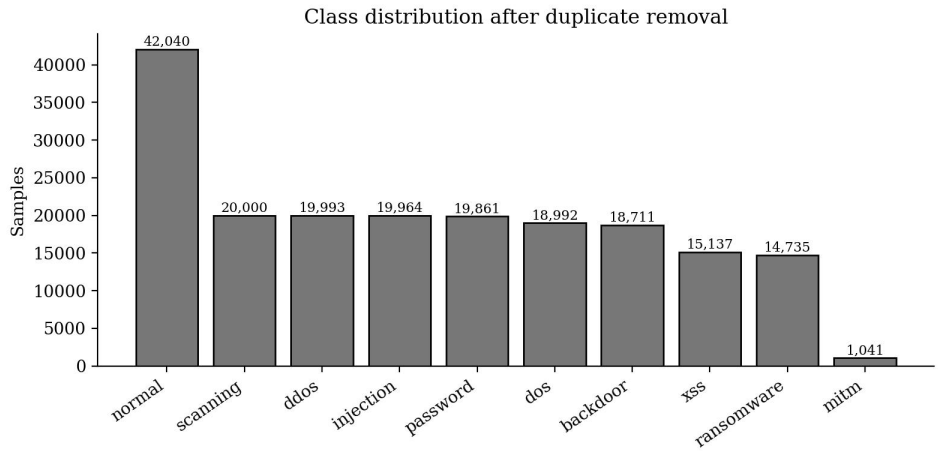


Figure 1. Class distribution after duplicate removal.

3.2 Numerical Features

Sixteen numerical features were selected: src_port, dst_port, duration, src_bytes, dst_bytes, missed_bytes, src_pkts, src_ip_bytes, dst_pkts, dst_ip_bytes, dns_qclass, dns_qtype, dns_rcode, http_request_body_len, http_response_body_len and http_status_code.

4. Assigned Pipeline and Methodology

4.1 Assigned Pipeline

The assigned pipeline is a high-level SOC workflow, shown below.

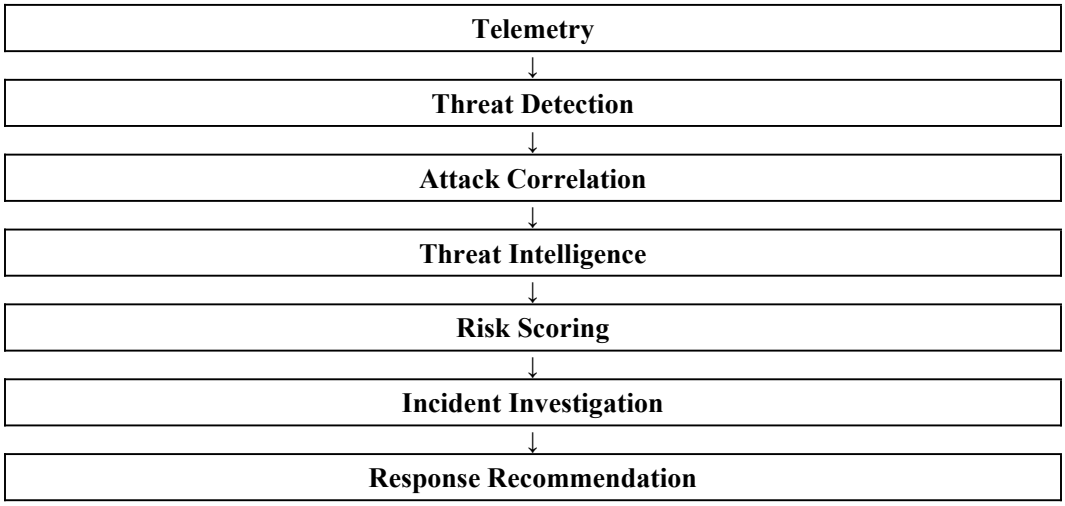


Figure 2. Assigned AutoSOC pipeline.

The assigned pipeline is broad. It does not say how noisy predictions are filtered, how alerts are related, or how correlation is validated, so these points were added in the updated version.

4.2 Methodology Flow

The notebook follows the assigned pipeline stage by stage. The final methodology is shown below.

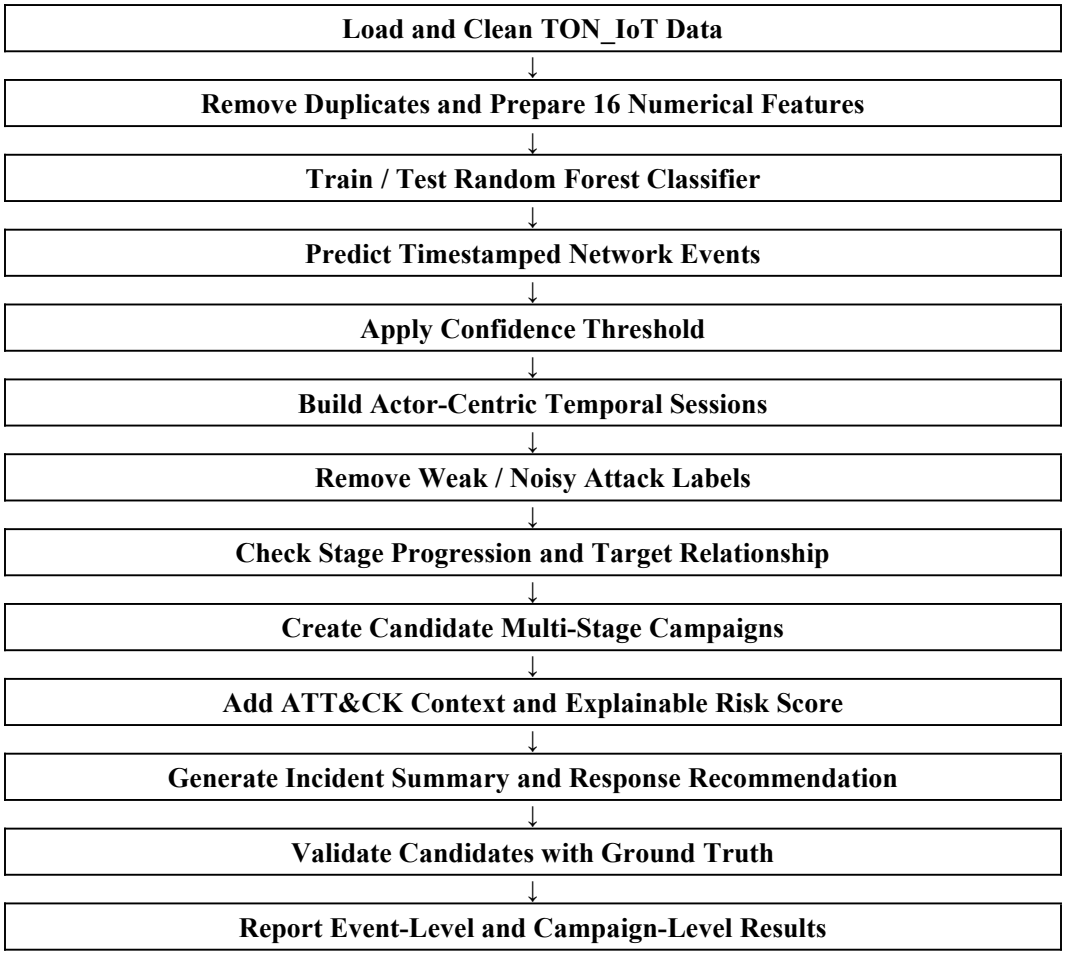


Figure 3. Final methodology used in the updated implementation.

5. Data Preprocessing

Exact duplicate rows were removed from the training dataset. Attack labels were converted to lowercase and extra spaces were removed. The sixteen selected traffic features were converted to numeric values. Invalid numerical values were converted to missing values and then filled with 0. The cleaned dataset was split into 80% training and 20% testing data using stratification and random_state = 42.

Only these 16 numerical columns were used. The categorical columns were left out to keep the model simple and explainable (see Limitations). The same numeric conversion was applied to every chunk of the external files.

6. Model Implementation

6.1 Random Forest Classifier

Setting	Value
Algorithm	sklearn RandomForestClassifier
Number of trees	150
Class weight	balanced_subsample
Random seed	42
Parallel jobs	All cores (n_jobs = -1)
Input	16 numerical features
Training / testing samples	152,379 / 38,095 (stratified 80/20)

Random Forest was chosen because it works well on numerical tabular data, needs no feature scaling, and gives feature importance directly. Class weighting was added because MITM has far fewer samples than the other classes.

6.2 Threat Detection and Confidence Filtering

Only non-normal predictions with confidence of at least 0.80 are sent to the correlation stage. Ground-truth labels are not used for this selection. The thresholds 0.70, 0.80 and 0.90 were compared, and 0.80 was kept as a practical baseline (Section 8.2).

6.3 Actor-Centric Temporal Correlation

The source IP is used as the candidate actor. Events from the same source are sorted by time, and a new campaign starts after an inactivity gap of more than 30 minutes. This lets one source be linked with several destinations in the same session.

6.4 Noise-Resistant Campaign Inference

A predicted attack type counts as a meaningful stage only if it has at least 10 events and at least 0.5% of the campaign events. A candidate multi-stage campaign needs two meaningful labels in two distinct stages, at least 75% forward stage transitions, a last stage later than the first, and shared targets in at least 50% of adjacent labels. A looser first version gave 14 candidates with weak validation, so these rules were added.

Parameter	Value	Purpose
Campaign inactivity window	1800 s	Split campaigns
Minimum events per attack label	10	Remove noise labels
Minimum share of campaign events	0.5%	Remove weak labels
Minimum forward progression	75%	Stages move forward
Minimum target relationship	50%	Shared targets

6.5 Attack Context and Stage Mapping

Predicted Attack Type	Analytical Stage
Scanning	Reconnaissance/Discovery
XSS	Initial Access
Injection	Initial Access
Password	Credential Access
MITM	Credential Access
Backdoor	Persistence
Ransomware	Impact
DoS	Impact
DDoS	Impact

The stage mapping is an analytical model for correlation, not proof of a complete MITRE ATT&CK kill chain.

6.6 Explainable Risk Scoring

The risk score (0 to 100) is used for prioritization, not as a probability of compromise. It has four parts: attack severity (maximum 40), event evidence (maximum 20), mean model confidence (maximum 20) and stage-progression evidence (maximum 20). Levels are LOW below 25, MEDIUM from 25 to below 50, HIGH from 50 to below 75, and CRITICAL at 75 or above.

6.7 Incident Investigation and Response Recommendation

For each campaign the system records the source actor, targets, attack and stage sequence, duration, events, confidence, timeline and risk explanation. Low and medium cases get monitoring advice, high and critical cases go to an analyst, and disruptive actions such as isolation or blocking are only recommended and need human approval.

6.8 Offline Ground-Truth Validation

Ground truth is used only after detection and correlation are finished, so there is no label leakage. Candidates are checked for label support, coverage, Jaccard similarity, ordered sequence support and exact match.

7. Experimental Setup and Evaluation Metrics

The system is evaluated at three levels:

- Held-out test: 20% random split of the training file (38,095 samples). Accuracy, precision, recall, F1 and the confusion matrix are reported.
- External files: separate timestamped files. Agreement, class-wise metrics, and agreement and coverage at three confidence thresholds are reported.
- Campaign level: candidates are compared with the ground-truth sequence of the same source IP and time window.

Precision = TP / (TP + FP), recall = TP / (TP + FN) and F1 = 2PR / (P + R). Macro averages give every class the same weight, and weighted averages use class size. Agreement is the share of retained events whose

predicted class matches the ground truth, and coverage is the share of all events above the confidence threshold.

For campaigns, predicted-label support is overlap divided by predicted labels, ground-truth coverage is overlap divided by ground-truth labels, and Jaccard is overlap divided by union. Ordered support and exact match compare the sequence order.

8. Experimental Results

8.1 Held-Out Model Performance

Metric	Result
Accuracy	98.74%
Macro Precision	0.965
Macro Recall	0.971
Macro F1	0.968
Weighted Precision	0.988
Weighted Recall	0.987
Weighted F1	0.987

Class	Precision	Recall	F1	Support
backdoor	1.00	1.00	1.00	3,742
ddos	0.99	0.98	0.98	3,999
dos	0.99	0.98	0.99	3,799
injection	0.98	0.97	0.97	3,993
mitm	0.76	0.83	0.79	208
normal	1.00	1.00	1.00	8,408
password	1.00	0.99	0.99	3,972
ransomware	1.00	1.00	1.00	2,947
scanning	0.99	0.99	0.99	4,000
xss	0.96	0.97	0.96	3,027

Table. Per-class results on the held-out test set.

This is the performance of the Random Forest on the random held-out split, not of the whole AutoSOC system. MITM is the weakest class (F1 0.79), which matches its small sample size.

Figure 4 shows that most mistakes are between injection and XSS: 87 injection samples were predicted as XSS and 56 XSS samples as injection.

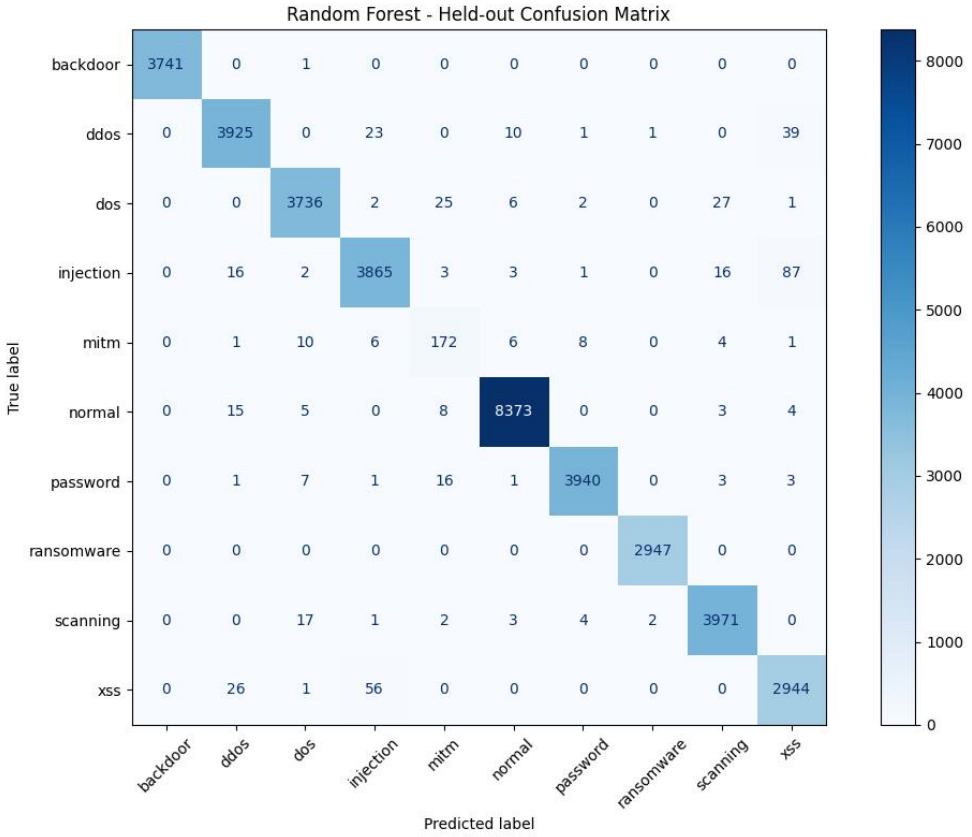


Figure 4. Random Forest held-out confusion matrix.

The strongest features are src_ip_bytes, dst_port, src_port, duration and dst_ip_bytes (Figure 5). The DNS fields have much lower importance.

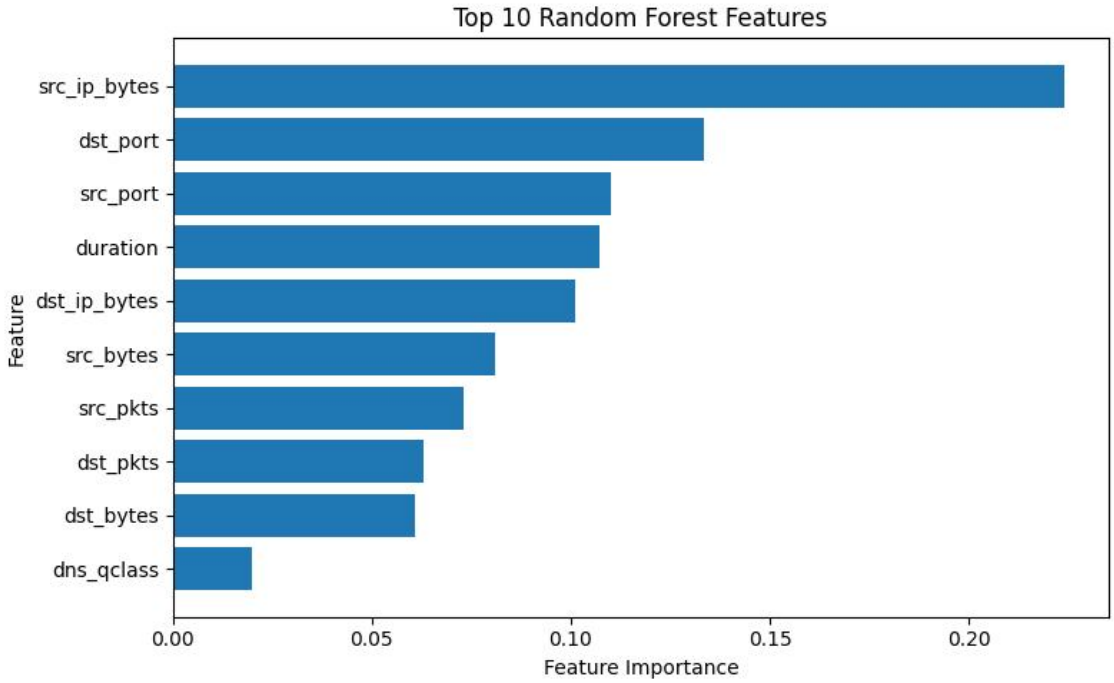


Figure 5. Top 10 Random Forest features.

8.2 External Evaluation and Threshold Comparison

On the separate timestamped files the overall prediction agreement was only 56.30%, much lower than the held-out result. Figure 6 shows agreement and coverage at three confidence thresholds. The 0.80 threshold was kept as the operational baseline (93.07% agreement with 41.57% coverage), which shows the usual trade-off between confidence and coverage.

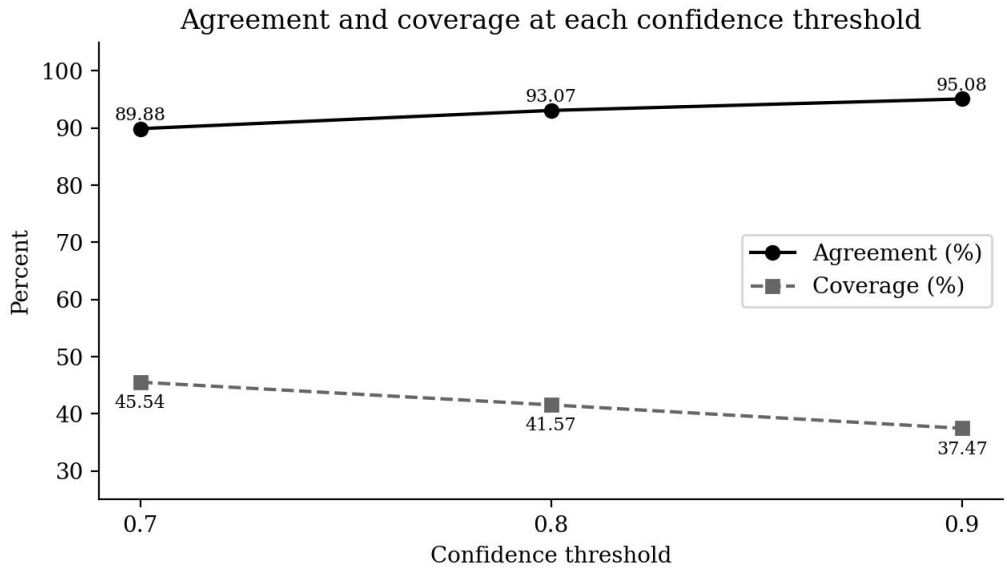


Figure 6. Agreement and coverage at different confidence thresholds.

Class	Precision	Recall	F1	Support
ddos	0.986	0.793	0.879	1,425,202
dos	0.152	0.023	0.040	975,261
normal	0.394	0.954	0.558	281,093
password	0.869	0.814	0.841	206,891
scanning	0.461	0.611	0.525	791,321
Other 5 classes	0.000	0.000	0.000	0

Table. Class-wise results on the external files.

The external files contain only five of the ten labels (ddos, dos, normal, password and scanning), so the metrics of the other five classes are 0.000 only because their support is 0. This is not a model error, but any of those five labels predicted on the external files is a wrong prediction. The weakest results are DoS (recall 0.023) and normal (precision 0.394).

8.3 Final Correlation Results

Metric	Final Result
High-confidence malicious events entering correlation	1,253,175
Total correlated actor-centric campaigns	2,576
Candidate multi-stage campaigns	4
Correlated multi-alert, non-progressive campaigns	15
Single-stage campaigns	2,557
Flow-to-campaign compression	99.79%

The 99.79% value is flow-to-campaign compression: many predicted malicious flows were grouped into a few campaign objects. It is not a measured reduction of real SOC alerts.

8.4 Final Candidate Multi-Stage Campaigns

Source IP	Targets	Predicted Sequence	Stage Sequence
192.168.1.31	5	XSS -> DDoS	Initial Access -> Impact
192.168.1.194	3	Scanning -> Ransomware	Reconnaissance/Discovery -> Impact
192.168.1.190	5	Scanning -> Ransomware	Reconnaissance/Discovery -> Impact
192.168.1.133	3	Scanning -> Ransomware	Reconnaissance/Discovery -> Impact

8.5 Ground-Truth Validation

Metric	Result
Candidates validated	4
Average predicted-label support	37.50%
Average ground-truth coverage	75.00%
Average Jaccard similarity	37.50%
Ordered sequence support	0
Exact sequence match	0

Three of the four candidates had partial ground-truth support (scanning was present, but the later ransomware stage was not confirmed), and the XSS -> DDoS candidate had no support in the matched window. No ordered or exact multi-stage sequence was confirmed. Ransomware has no ground-truth events in the external files, so a ransomware stage predicted there is most likely a classifier error.

8.6 Risk Results for the Four Candidates

Candidate	Risk Score	Risk Level	Validation Status
XSS -> DDoS	91.73	CRITICAL	No ground-truth attack support
Scanning -> Ransomware	86.06	CRITICAL	Partial support
Scanning -> Ransomware	85.52	CRITICAL	Partial support
Scanning -> Ransomware	83.15	CRITICAL	Partial support

The risk score is a heuristic priority score based on predicted evidence. A CRITICAL level does not mean the attack was confirmed by ground truth.

9. Suggestions Collected from Five AI Systems

After the first implementation, the project description and the notebook were reviewed with five AI systems (ChatGPT, Gemini, Claude, Qwen and Grok). Only the suggestions that suited this project were selected.

AI System	Main Suggestions
ChatGPT	Data-quality validation, confidence handling, entity and temporal correlation, explainable risk scoring.
Gemini	Stronger preprocessing, multi-entity correlation, non-linear stages, tiered response.
Claude	External-file performance, confidence calibration, actor-centric correlation, campaign validation.
Qwen	Keep confidence filtering, define temporal correlation, add explainability, ATT&CK as context.
Grok	Keep lightweight Random Forest, explicit confidence and temporal rules, candidate terminology.

10. Final Decisions from the AI Suggestions

Suggestion	Decision	Reason
Data-quality validation	Accepted	It improves input reliability and is easy to explain.
Confidence-aware filtering	Accepted	It prevents many uncertain predictions from entering correlation.
Actor-centric / entity-based correlation	Accepted	It can connect one source actor with several related targets.
Flexible stage progression	Accepted	A real sequence may contain repeated or intermediate stages.
MITRE ATT&CK context	Accepted as context	Static mapping is useful for explanation but is not a full threat-intelligence system.
Human approval for disruptive response	Accepted	The prototype recommends actions but does not execute isolation or blocking.
Binary model before multi-class model	Not used	It would require another model and was not necessary for the current objective.
NetworkX/GNN attack graph	Future work	Useful, but unnecessary for the current Colab implementation.

11. Proposed Updated AutoSOC Pipeline

The updated pipeline keeps the assigned workflow and adds the technical steps that were missing from it.

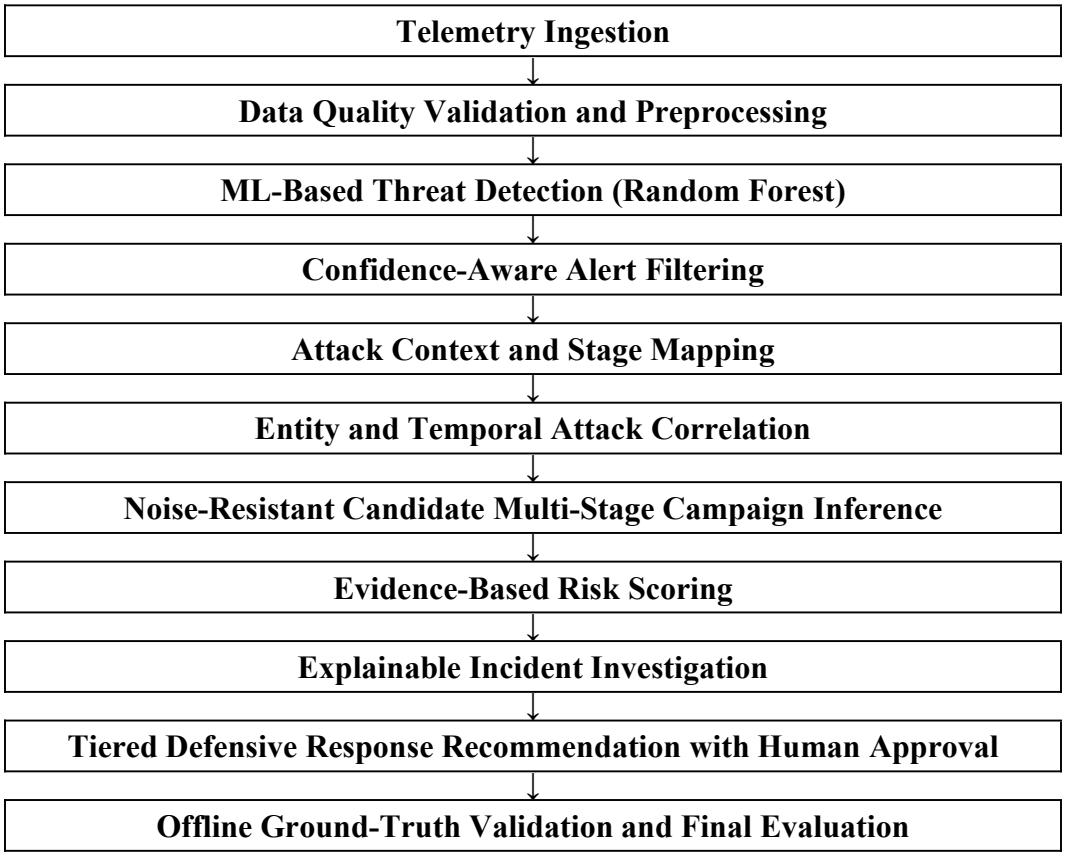


Figure 7. Final updated AutoSOC pipeline developed after reviewing the five AI suggestions.

12. Comparison: Old Pipeline vs Updated Pipeline

Area	Old Pipeline	Updated Pipeline	AI Systems Behind It
Telemetry	Telemetry used as it is	Telemetry plus duplicate removal, label cleaning and numeric conversion	ChatGPT, Gemini
Threat Detection	Random Forest prediction, no rule for noisy predictions	Random Forest plus confidence filter (0.80) and threshold comparison	ChatGPT, Claude, Qwen, Grok
Attack Correlation	Source IP + destination IP, 30-minute window	Actor-centric sessions by source IP, 30-minute inactivity window	ChatGPT, Gemini, Claude, Qwen, Grok
Campaign Inference	Strict forward stage order	Minimum label support, 75% progression, first-to-last check, 50% target relationship	ChatGPT, Gemini, Claude, Grok
Threat Intelligence	Not defined in detail	ATT&CK context and stage mapping, used as context only	Gemini, Qwen, Grok
Risk Scoring	One score, no breakdown	Four visible parts, total 100	ChatGPT, Claude, Qwen
Incident Investigation	Basic summary	Timeline, evidence and risk explanation	Claude, Qwen
Response	One recommendation	Tiered recommendation, analyst approval for disruptive actions	Gemini
Evaluation	Held-out accuracy only	Held-out, external, threshold and campaign-level validation	ChatGPT, Claude, Qwen, Grok

13. Improvement Through Iteration

The first implementation followed the assigned pipeline and grouped events by source IP, destination IP and a 30-minute window. The first AI-updated actor-centric version gave 14 candidates but weak validation, so minimum label support, 75% progression, a first-to-last forward check and 50% target relationship were added. The final version has fewer but better-supported candidates.

Version	Campaigns	Candidate Multi-stage	Avg. Jaccard	GT Coverage
Original final baseline	3,330	2	16.66%	50.00%
First AI-updated actor-centric version	2,576	14	16.55%	50.00%
Final refined updated version	2,576	4	37.50%	75.00%

14. Discussion

The updated pipeline is stronger than the assigned high-level pipeline because every main decision is visible: data cleaning, confidence handling, event grouping, noise removal, stage checking and validation.

The external agreement (56.30%) is much lower than the held-out accuracy (98.74%), which shows that a random split can be too optimistic for separate timestamped files. Four campaigns passed the final rules, but none had a complete ordered ground-truth sequence, so the result is candidate campaign inference, not confirmed attack reconstruction.

15. Limitations

- The model shows a clear generalization gap between the random held-out split and the separate timestamped network files.
- The external files contain only five of the ten classes, so the metrics of the other five classes are 0.000 because their support is 0.
- Only 16 numerical features were used. The categorical columns of the dataset were not used and may contain useful information.
- The correlation engine is rule-based. Temporal proximity, a common source, and target overlap provide evidence of relation but do not prove common causality.
- The risk score is a prototype priority score with manually selected components and thresholds.
- Ground-truth coverage may be incomplete, so absence of support should be described as lack of available ground-truth evidence rather than definite proof that an event did not occur.

16. Conclusion

AutoSOC first implemented the assigned pipeline and then improved it with explicit preprocessing, confidence-aware filtering, actor-centric temporal correlation, noise-resistant campaign inference, explainable risk scoring and offline ground-truth validation. The Random Forest reached 98.74% held-out accuracy, but only 56.30% agreement on the separate files (93.07% for events with confidence of at least 0.80, at 41.57% coverage). The final stage grouped 1,253,175 events into 2,576 campaigns and produced four candidate multi-stage campaigns, with Jaccard similarity improved from 16.55% to 37.50% and ground-truth coverage to 75.00%. No complete ordered attack chain was confirmed, so the results are reported as candidate campaigns. Future work can calibrate the probabilities and add the categorical features.